

ALEKSANDAR STOJKOSKI

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SUMMARY

Third-year Mechanical Engineering (Honours) student with hands-on experience delivering client-focused mechanical and structural design solutions using ABAQUS and SPACE GASS. Skilled in stakeholder engagement, finite element analysis and translating complex technical problems into validated engineering outcomes. Curious, collaborative, and motivated to bring a multidisciplinary mindset to real technical and advisory work.

EDUCATION

Bachelor of Engineering (Honours) - Mechanical

Deakin University, Burwood

Expected 2027

Relevant coursework: Control Systems Engineering, Dynamics of Machines, Thermo-Fluid Systems, Product Design

ENGINEERING PROJECTS

Design Engineering Project - Universal Chain Guide Attachment (SpiderGuide) 2026

- Designed the "SpiderGuide", a universal chain guide attachment for electric chain hoists, resolving a chain-spooling fault in Melbourne Park's live entertainment rigging system as part of a real client brief.
- Benchmarked five competing products and translated stakeholder requirements into design criteria (tool-free installation, non-invasive mounting, sub-5 kg weight), delivering a final design under the 5 kg target and ready for client evaluation.
- Led ABAQUS FEA (static, frequency, buckling) across the AL6061 sheet-metal assembly, confirming no plastic deformation in any component and identifying the clamp bracket as the governing member (FOS 2.37x, vs. up to 4,940x elsewhere) to pinpoint where design margin mattered most.

Formula SAE Team Establishment Initiative - Deakin University 2026 – Present

- Co-leading the proposal to establish Deakin Burwood's first Formula SAE program, working with the university's motorsport program lead to turn the concept into a structured, self-sufficient implementation plan.
- Navigating competing resourcing priorities and stakeholder constraints across Deakin's Burwood and Geelong campuses to build a staged proposal with realistic milestones.
- Preparing a business case and executive presentation for senior engineering leadership, evaluating program feasibility and demonstrating its value to student outcomes, industry engagement and university growth.

Structural Design Project - Cyclist & Pedestrian Bridge 2025

- Developed the preliminary structural design of a cable-stayed cyclist and pedestrian bridge for the City of Stonnington, analysing the main girder, A-frame tower, stay cables and footing using SPACE GASS and independent hand calculations to verify structural performance.
- Responded to revised geotechnical constraints by redesigning the footing for a reduced allowable bearing pressure of 100 kPa, maintaining compliance with bearing ($FOS \geq 3.0$) and sliding/overturning ($FOS \geq 1.5$) requirements without redesigning the entire structure.

SKILLS

Technical Skills

Structural & Analysis

- SPACE GASS
- ABAQUS FEA

CAD

- Fusion 360

Programming

- Python
- MATLAB
- AI-assisted workflows (Claude, ChatGPT)

Soft Skills

- Clear communication and technical presentation
- Analytical problem-solving
- Teamwork and collaboration
- Stakeholder engagement
- Time management and adaptability

PROFESSIONAL EXPERIENCE

Retail Athlete (Sales Assistant) - Nike

2022 – Present

- Provided customer service in a high-volume retail environment, meeting sales and performance targets while balancing full-time university study.

VOLUNTEERING & ACTIVITIES

Deakin Engineering Ambassador - Burwood Campus

2024 – Present

- Selected by the Head of Engineering in first year to represent Deakin's inaugural Burwood engineering cohort, presenting student projects to politicians, council representatives and industry stakeholders at a major campus promotion event.
- Supporting ongoing advocacy for engineering at Deakin through orientation and outreach events.

Track Marshal - Preston Motorcycle Club, Broadford

2024 – Present

- Safety-critical trackside role requiring sustained situational awareness and rapid incident response.